

Solving linear dynamic systems

From difference equations to Blanchard–Kahn

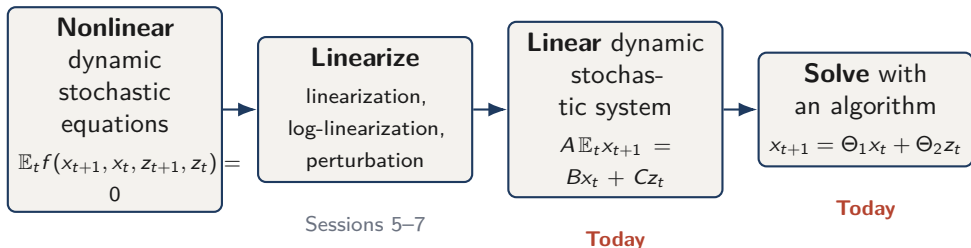
Computational Methods for Macroeconomics — Sessions 2–3

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Last time: a model is a set of equations.

This time: when those equations are linear, dynamic, and stochastic,
how do we actually solve them?

The lifecycle of a DSGE model solution



- The equations of a DSGE model are nonlinear. Almost every solution method first replaces them with **linear** equations near a steady state.
- We will *not* do the linearization step today — that is for later sessions.
- Today we take the linear system as given and ask: **what does it mean to solve it, and when does a solution exist and is unique?**

The algorithms: a map of the territory

Method	Key tool	Handles
Blanchard–Kahn (1980) (Jordan variant)	eigen-decomposition of $A^{-1}B$ Jordan form	A invertible, diagonalizable A invertible, not diagonalizable
Klein (2000)	generalized Schur (QZ)	A singular
Sims (2002), gensys	QZ + SVD	A singular; no need to label predetermined variables
Uhlig (1999)	undetermined coefficients	“toolkit” form; solves a matrix quadratic equation
Sequence space (Auclert et al. 2021)	stacked Jacobians	no state-space form at all; solves for whole time paths

All the *state-space* methods are one idea in different clothes: **separate stable from unstable directions; solve the unstable ones forward, the stable ones backward.**

Plan for these two sessions

Deterministic (Miao ch. 1)

1. Scalar first-order equations
2. Lag operators
3. Scalar second-order equations
4. First-order linear *systems*
⇒ Blanchard–Kahn
5. Beyond BK: Jordan, Klein, Sims
6. Phase diagrams

Stochastic (Miao ch. 2)

7. Shocks, stationarity, expectations
8. Lag operators with $\mathbb{E}_t$
9. Scalar RE models: forward solution, undetermined coefficients
10. Multivariate RE models: BK again
11. A New Keynesian example

Two questions run through everything: **Is the initial value given?** and **Is the coefficient (eigenvalue) inside or outside the unit circle?**

The simplest dynamic equation

$$x_{t+1} = b x_t + c z_t, \quad t \geq 0$$

- x_t , b , c , z_t are real numbers; $\{z_t\}$ is an exogenously given *bounded* sequence.
- **Autonomous**: z_t is constant for all t . **Homogeneous**: $c z_t = 0$ for all t .

What does “solve” mean here?

Find the whole sequence $\{x_t\}_{t \geq 0}$ that satisfies the equation for every t . Whether we can — and how many ways we can — depends on two things:

1. Is x_0 given to us?
2. Is $|b| < 1$ or $|b| > 1$?

[Miao §1.1, eq. (1.1)]

The autonomous case: steady state and general solution

Set $z_t = 1$ (no loss of generality): $x_{t+1} = bx_t + c$.

- A **steady state** (stationary point) is a constant solution $x_t = \bar{x}$:

$$\bar{x} = b\bar{x} + c \implies \bar{x} = \frac{c}{1-b}, \quad b \neq 1.$$

- Subtract $\bar{x}$ from both sides: $x_{t+1} - \bar{x} = b(x_t - \bar{x})$. Deviations from the steady state follow a homogeneous equation, so

$$x_t = (x_0 - \bar{x}) b^t + \bar{x}. \quad (\text{general solution})$$

Long-run behaviour: it all depends on $|b|$

$$x_t = (x_0 - \bar{x}) b^t + \bar{x}$$

Consider when $t \rightarrow \infty$. Two cases:

$$|b| < 1$$

$b^t \rightarrow 0$: the solution converges to $\bar{x}$ for *any* x_0 .

$\bar{x}$ is **globally asymptotically stable**.

But if x_0 is *not* given, *every* x_0 gives a valid, convergent path: the solution is **indeterminate**.

$$|b| > 1$$

b^t explodes: the solution is **unstable** for any $x_0 \neq \bar{x}$.

Even if x_0 is not given there is exactly *one solution*: $x_t = \bar{x}$ for all t .

Instability + freedom to choose $x_0 \Rightarrow$ **uniqueness**.

This is an important point: keep it in mind when we get to eigenvalues.

Non-autonomous case I: x_0 given $\Rightarrow$ solve backward

$x_{t+1} = bx_t + cz_t$ with x_0 exogenously given. Substitute repeatedly:

$$x_1 = bx_0 + cz_0$$

$$x_2 = bx_1 + cz_1 = b^2x_0 + c(bz_0 + z_1)$$

$$\vdots$$

$$x_t = b^t x_0 + c \sum_{j=0}^{t-1} b^j z_{t-1-j} \quad (\text{backward-looking solution})$$

- x_t is a weighted sum of *past* shocks $z_0, \dots, z_{t-1}$ plus the initial condition.
- If $|b| < 1$: $b^t x_0 \rightarrow 0$ and the sum converges (since $\{z_t\}$ is bounded). The limit is a **generalized steady state**; it does not depend on x_0 .
- If $|b| > 1$: $b^t x_0$ explodes — the path diverges.

[Miao eq. (1.4)–(1.5)]

Non-autonomous case II: x_0 not given $\Rightarrow$ solve forward

Now suppose x_0 is *not* given (think: x_t is an asset price). Rewrite $x_t = \frac{1}{b}x_{t+1} - \frac{c}{b}z_t$ and substitute *forward*:

$$\begin{aligned}x_t &= \frac{1}{b}x_{t+1} - \frac{c}{b}z_t = \frac{1}{b}\left(\frac{1}{b}x_{t+2} - \frac{c}{b}z_{t+1}\right) - \frac{c}{b}z_t \\&= \left(\frac{1}{b}\right)^T x_{t+T} - \frac{c}{b} \sum_{j=0}^{T-1} \left(\frac{1}{b}\right)^j z_{t+j} \quad \text{for any } T \geq 1.\end{aligned}$$

- x_t now depends on *future* z 's — and on a terminal term $b^{-T}x_{t+T}$.
- To let $T \rightarrow \infty$ we need to say something about that terminal term.

[Miao eq. (1.6)]

The transversality condition

Transversality (no-bubble) condition

$$\lim_{T \rightarrow \infty} \left(\frac{1}{b}\right)^T x_{t+T} = 0.$$

Imposing it, $T \rightarrow \infty$ gives the **forward-looking solution**:

$$x_t = -\frac{c}{b} \sum_{j=0}^{\infty} \left(\frac{1}{b}\right)^j z_{t+j}.$$

- If $|b| > 1$ the sum converges ($\{z_t\}$ bounded) and satisfies the TVC itself; the solution is **stable** (bounded).
- Economics: x_t cannot grow at rate b forever — an optimality condition in optimization problems, a no-bubble condition in asset pricing.
- Without it, $x_t = x_t^* + B_t$, $B_{t+1} = bB_t$, also solves the equation for any B_0 [Mitra eq. (1.7)–(1.9)]

Where we stand: two conditions decide everything

	$ b < 1$	$ b > 1$
x_0 given	backward solution, converges	backward solution, explodes
x_0 not given	indeterminate (any x_0 works)	unique stable forward solution

- Whether x_0 is given will become: is the variable **predetermined**?
- Whether $|b| \leq 1$ will become: is the **eigenvalue** stable or unstable?
- The happy diagonal — predetermined with stable root, free with unstable root — is exactly what the Blanchard–Kahn conditions will demand, one root per variable.

The lag operator

For a sequence $\{X_t\}_{t=-\infty}^{\infty}$, define the **lag operator** L by

$$LX_t = X_{t-1}, \quad L^n X_t = X_{t-n} \quad \text{for all integers } n.$$

- n can be negative: $L^{-1}X_t = X_{t+1}$ is the **lead** (forward) operator.
- $L^0 X_t = X_t$, and $L^m L^n = L^{m+n}$.
- $L^n c = c$ for any constant c that does not depend on time.

Why bother?

L lets us treat a difference equation as an *algebraic* equation in the operator, and manipulate it — factorize, invert, expand — like a polynomial.

[Miao §1.2]

Properties we will use

1. **Linearity:** $L(aX_t + bY_t) = aX_{t-1} + bY_{t-1}$.
2. **Polynomials in L :** $(1 - \lambda L)X_t = X_t - \lambda X_{t-1}$; so $x_{t+1} = bx_t + cz_t$ reads $(L^{-1} - b)x_t = cz_t$.
3. **Polynomials commute and factorize** like ordinary polynomials:
 $(L^{-1} - \lambda_1)(L^{-1} - \lambda_2) = L^{-2} - (\lambda_1 + \lambda_2)L^{-1} + \lambda_1\lambda_2$.
4. **Geometric expansion** (main property). For $|\lambda| < 1$,

$$\frac{1}{1 - \lambda L^n} = \sum_{j=0}^{\infty} \lambda^j L^{nj}$$

5. **Matrix version:** $(I - AL^n)^{-1} = \sum_{j=0}^{\infty} A^j L^{nj}$ if all eigenvalues of A are inside the unit circle.

The two solutions of §1.1, in one line each

$$x_{t+1} = bx_t + cz_t, \text{ i.e. } (L^{-1} - b)x_t = cz_t, \text{ i.e. } (1 - bL)x_{t+1} = cz_t.$$

Backward ($|b| < 1$): expand in L

$$x_{t+1} = \frac{cz_t}{1 - bL} = c \sum_{j=0}^{\infty} b^j L^j z_t = c \sum_{j=0}^{\infty} b^j z_{t-j}.$$

Forward ($|b| > 1$): factor out $-b$ and expand in L^{-1}

$$x_t = \frac{cz_t}{L^{-1} - b} = \frac{cz_t}{-b(1 - b^{-1}L^{-1})} = -\frac{c}{b} \sum_{j=0}^{\infty} b^{-j} L^{-j} z_t = -\frac{c}{b} \sum_{j=0}^{\infty} b^{-j} z_{t+j}.$$

The direction of expansion is *forced* by which of b , $1/b$ is inside the unit circle.

A second-order equation

$$x_{t+2} = a x_{t+1} + b x_t + c z_t, \quad x_0 \text{ given,}$$

a, b, c real constants, $\{z_t\}$ bounded. In operator form:

$$(L^{-2} - aL^{-1} - b) x_t = c z_t.$$

Characteristic equation

$$\lambda^2 - a\lambda - b = 0, \quad \text{roots } \lambda_1, \lambda_2 \text{ with } \lambda_1 + \lambda_2 = a, \lambda_1 \lambda_2 = -b.$$

So the operator polynomial **factorizes**:

$$(L^{-1} - \lambda_1)(L^{-1} - \lambda_2) x_t = c z_t.$$

[Miao §1.3, eq. (1.13)–(1.15)]

Homogeneous solution: real, distinct roots

Solution = (any particular solution) + (general solution of the homogeneous equation $x_{t+2} = ax_{t+1} + bx_t$). For real distinct roots:

$$x_t^h = c_1 \lambda_1^t + c_2 \lambda_2^t,$$

with c_1, c_2 pinned down by *two* boundary conditions. We have only one (x_0).

Roots	Steady state is a...	What happens
$ \lambda_1 , \lambda_2 < 1$	sink	every path converges; c_1, c_2 free $\Rightarrow$ indeterminate
$ \lambda_1 , \lambda_2 > 1$	source	every path explodes (unless x starts at $\bar{x}$)
$ \lambda_1 < 1 < \lambda_2 $	saddle	exactly one convergent path: the saddle path

The saddle is the interesting case — and the typical case in macro models.

Solving the saddle case with lag operators

Assume $|\lambda_1| < 1 < |\lambda_2|$. From the factorized equation,

$$(L^{-1} - \lambda_1) x_t = \frac{c z_t}{L^{-1} - \lambda_2} = \frac{-c z_t}{\lambda_2 (1 - \lambda_2^{-1} L^{-1})} = -\frac{c}{\lambda_2} \sum_{j=0}^{\infty} \lambda_2^{-j} z_{t+j},$$

expanding *forward* because $|\lambda_2^{-1}| < 1$. Hence

$$x_{t+1} = \lambda_1 x_t - \frac{c}{\lambda_2} \sum_{j=0}^{\infty} \lambda_2^{-j} z_{t+j}$$

- The unstable root λ_2 was “solved forward” and disappeared into a discounted sum of future z ’s.
- What remains is a *first-order* equation with the stable root λ_1 , solved *backward* from the given x_0 . One stable root $\leftrightarrow$ one initial condition; one unstable root $\leftrightarrow$ one free variable.

[Miao eq. (1.18)]

From a second-order equation to a first-order system

$x_{t+2} = ax_{t+1} + bx_t + cz_t$ can be stacked:

$$\begin{bmatrix} x_{t+1} \\ x_{t+2} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ b & a \end{bmatrix} \begin{bmatrix} x_t \\ x_{t+1} \end{bmatrix} + \begin{bmatrix} 0 \\ c \end{bmatrix} z_t.$$

- x_t is **predetermined** (its initial value x_0 is given); x_{t+1} is **non-predetermined**.
- Any higher-order equation can be stacked into a first-order system the same way. So we only ever need to study first-order systems.

The general first-order linear system

$$A x_{t+1} = B x_t + C z_t, \quad t = 0, 1, 2, \dots$$

$x_t \in \mathbb{R}^n$ endogenous, $z_t \in \mathbb{R}^{n_z}$ exogenous, A , B , C conformable matrices.

[Miao §1.4, eq. (1.19)–(1.20)]

Vocabulary: steady state and stability

- **Steady state** $\bar{x}$: a point such that once $x_t = \bar{x}$, $x_s = \bar{x}$ for all $s > t$. If $z_t = \bar{z}$ eventually and $A - B$ is invertible, then $\bar{x} = (A - B)^{-1}C\bar{z}$.
- A sequence $\{X_t\}$ is **stable** if it is *bounded*: there is $M > 0$ with $\|X_t\|_{\max} < M$ for all t . (Growth is allowed after detrending.)
- $\bar{X}$ is **asymptotically stable** for $\{X_t\}$ if $X_t \rightarrow \bar{X}$ for some X_0 ; **globally asymptotically stable** if for *any* X_0 .

Standing assumption

$\{z_t\}$ is a stable (bounded) sequence. Our goal is to find a **stable solution** $\{x_t\}$ of $Ax_{t+1} = Bx_t + Cz_t$.

We skip Miao's regularity assumption $\det(A\alpha - B) \neq 0$ for some α ; it is automatically satisfied when A is invertible, which we assume next. [Miao Def. 1.4.1–1.4.4, Assumption 1.4.3]

The nonsingular case: $W = A^{-1}B$

Assume A is invertible. Premultiply by A^{-1} : $x_{t+1} = Wx_t + A^{-1}Cz_t$, where $W \equiv A^{-1}B$.

Definition: diagonalizable

An $n \times n$ matrix W is **diagonalizable** if it has n linearly independent eigenvectors. Equivalently, there is an invertible P and a diagonal D with

$$W = P D P^{-1}, \quad D = \text{diag}(\lambda_1, \dots, \lambda_n),$$

λ_i the eigenvalues of W ($\det(W - \lambda I) = 0$), column i of P the eigenvector for λ_i ($WP = PD$).

- An eigenvalue is **stable** if $|\lambda| < 1$, **unstable** if $|\lambda| > 1$.
- Assume for now: W is **diagonalizable** with **no eigenvalue on the unit circle**. (Jordan form relaxes the first assumption — later.)

[Miao §1.4.1, Def. 1.4.5; Miao writes $W = P^{-1}JP$, so his P is our P^{-1}]

Decoupling the system

Define transformed variables and shocks:

$$x_t^* \equiv P^{-1}x_t, \quad C^* \equiv P^{-1}A^{-1}C.$$

Then $x_{t+1} = PDP^{-1}x_t + A^{-1}Cz_t$ becomes, after premultiplying by P^{-1} ,

$$x_{t+1}^* = Dx_t^* + C^*z_t$$

- Because D is diagonal, this is n separate scalar equations,

$$x_{i,t+1}^* = \lambda_i x_{i,t}^* + C_i^* z_t, \quad i = 1, \dots, n,$$

each of which we already know how to solve (§1.1)!

- Each one is solved backward if $|\lambda_i| < 1$ and forward if $|\lambda_i| > 1$.
- The only difficulty: the initial conditions we are given are on x_t , not on x_t^* [Miao eq. (1.23)–(1.24)]

When all of x_0 is given: the backward solution

If the whole vector x_0 is given, $x_0^* = P^{-1}x_0$ is known, and solving backward

$$x_t^* = D^t x_0^* + \sum_{j=0}^{t-1} D^j C^* z_{t-1-j}, \quad x_t = P x_t^*.$$

Theorem (sink)

Suppose A and $I - W$ are nonsingular and $z_t = \bar{z}$ for $t \geq T$. Then $x_t \rightarrow \bar{x} = (A - B)^{-1} C \bar{z}$ for *any* x_0 if and only if **all eigenvalues of W are stable**.

- Proof idea: $D^t \rightarrow 0$ iff every $|\lambda_i| < 1$; then $x_t^* \rightarrow (I - D)^{-1} C^* \bar{z}$ and $x_t \rightarrow P(I - D)^{-1} P^{-1} A^{-1} C \bar{z} = (I - W)^{-1} A^{-1} C \bar{z} = \bar{x}$.
- All eigenvalues stable: **sink**. All unstable: **source**.
- But in economics, part of x_0 is typically *not* given. . .

[Miao Theorem 1.4.1]

Predetermined and non-predetermined variables

Definition (deterministic case)

A variable is **predetermined** if its initial value is exogenously given. Otherwise it is **non-predetermined**, a **jump** variable.

Order the vector so that predetermined variables come first:

$$x_t = \begin{bmatrix} k_t \\ y_t \end{bmatrix}, \quad \begin{array}{ll} k_t : & n_k \text{ predetermined variables (capital, debt, lagged prices, ...)} \\ y_t : & n_y \text{ non-predetermined variables (consumption, inflation, asset prices, ...)} \end{array}$$

with $n_k + n_y = n$. Likewise count the eigenvalues of W : n_s stable, n_u unstable, $n_s + n_u = n$ (none on the unit circle).

Partitioning

Order the eigenvalues in D with the stable ones first, and partition everything conformably:

$$D = \begin{bmatrix} D_s & 0 \\ 0 & D_u \end{bmatrix}, \quad x_t^* = \begin{bmatrix} s_t \\ u_t \end{bmatrix}, \quad C^* = \begin{bmatrix} C_s^* \\ C_u^* \end{bmatrix},$$

where D_s is $n_s \times n_s$ (all $|\lambda| < 1$) and D_u is $n_u \times n_u$ (all $|\lambda| > 1$). The decoupled system is

$$\begin{bmatrix} s_{t+1} \\ u_{t+1} \end{bmatrix} = \begin{bmatrix} D_s & 0 \\ 0 & D_u \end{bmatrix} \begin{bmatrix} s_t \\ u_t \end{bmatrix} + \begin{bmatrix} C_s^* \\ C_u^* \end{bmatrix} z_t.$$

The link between original and transformed variables, $x_t = Px_t^*$ and $x_t^* = P^{-1}x_t$:

$$\begin{bmatrix} k_t \\ y_t \end{bmatrix} = \begin{bmatrix} P_{ks} & P_{ku} \\ P_{ys} & P_{yu} \end{bmatrix} \begin{bmatrix} s_t \\ u_t \end{bmatrix}, \quad \begin{bmatrix} s_t \\ u_t \end{bmatrix} = \begin{bmatrix} M_{sk} & M_{sy} \\ M_{uk} & M_{uy} \end{bmatrix} \begin{bmatrix} k_t \\ y_t \end{bmatrix}, \quad M \equiv P^{-1}.$$

E.g. P_{ks} is $n_k \times n_s$ and M_{uy} is $n_u \times n_y$.

[Miao eq. (1.26)–(1.28); Miao's P is our M , his R is our P]

The Blanchard–Kahn theorem

Theorem (Blanchard and Kahn, 1980)

Suppose

- (i) the number of unstable eigenvalues equals the number of non-predetermined variables: $n_u = n_y$ (order / counting condition)
- (ii) M_{uy} is nonsingular (rank condition)
- (iii) W has no eigenvalue on the unit circle. (hyperbolicity)

Then the system has a **unique stable solution** for any given k_0 and any stable $\{z_t\}$.

The proof is constructive: it *is* the solution algorithm. Three steps:

1. Solve the unstable block u_t **forward**.
2. Recover the jump variables y_t from u_t and k_t (needs the rank condition).
3. Solve the stable block s_t — hence k_t — **backward** from k_0 .

[Miao Theorem 1.4.2]

Proof, step 1: the unstable block goes forward

The lower block reads $u_{t+1} = D_u u_t + C_u^* z_t$, i.e. $(L^{-1} - D_u) u_t = C_u^* z_t$. Since every eigenvalue in D_u exceeds one in modulus, D_u^{-1} is a stable matrix, so we expand *forward*:

$$\begin{aligned} u_t &= (L^{-1} - D_u)^{-1} C_u^* z_t = -D_u^{-1} (I - D_u^{-1} L^{-1})^{-1} C_u^* z_t \\ &= -\sum_{j=0}^{\infty} D_u^{-j-1} C_u^* z_{t+j}. \end{aligned}$$

- Converges because $\{z_t\}$ is stable. This is the only stable solution for u_t (any other one adds an exploding D_u^t term — the TVC logic of §1.1).
- **u_t is now completely determined by exogenous variables**, for every t including $t = 0$.
- u_t is an n_u -vector. We have used up n_u “degrees of freedom”.

[Miao eq. (1.29)]

Proof, step 2: recover the jump variables

From $x_t^* = Mx_t$, the lower block is

$$u_t = M_{uk} k_t + M_{uy} y_t.$$

- M_{uy} is $n_u \times n_y$. If $n_u = n_y$ it is square, and if it is nonsingular,

$$y_t = M_{uy}^{-1}(u_t - M_{uk}k_t)$$

- Given k_t (from step 3) and u_t (from step 1), the jump variables are pinned down.
At $t = 0$: $y_0 = M_{uy}^{-1}(u_0 - M_{uk}k_0)$ — the “missing” initial condition!

This is where the two BK conditions bite

n_u equations (the forward solution fixed u_t) in n_y unknowns (y_t). A unique solution needs as many independent equations as unknowns: $n_u = n_y$ and M_{uy} invertible. Session 1’s “solvable”, again.

[Miao eq. (1.31)–(1.32)]

Proof, step 3: the stable block goes backward

The upper block $s_{t+1} = D_s s_t + C_s^* z_t$ is stable, so we solve it backward — once we know s_0 . From $x_t = P x_t^*$, the upper block is $k_t = P_{ks} s_t + P_{ku} u_t$.

- Under (i) P_{ks} is square ($n_k = n_s$), and one can show it is invertible exactly when M_{uy} is (block-inverse formula: $P_{ks} = (M_{sk} - M_{sy} M_{uy}^{-1} M_{uk})^{-1}$). So

$$s_0 = P_{ks}^{-1} (k_0 - P_{ku} u_0).$$

- Then iterate $s_{t+1} = D_s s_t + C_s^* z_t$: a bounded sequence, since D_s is stable.
- Finally map back: $k_t = P_{ks} s_t + P_{ku} u_t$ and $y_t = P_{ys} s_t + P_{yu} u_t$.

Result

$\{x_t\}$ is stable; it satisfies the system for every t and starts at the given k_0 . Uniqueness: any other stable solution must share u_t (step 1), hence y_t (step 2), hence s_t (step 3).



[Miao eq. (1.33) gives the k_t recursion directly]

The algorithm, in one box

Blanchard–Kahn algorithm for $x_{t+1} = Wx_t + A^{-1}Cz_t$

1. Compute $W = A^{-1}B$, its eigenvalues and eigenvectors: $W = PDP^{-1}$.
2. Sort: stable eigenvalues first. Count n_u ; compare with n_y . Check M_{uy} invertible.
3. $u_t = -\sum_{j \geq 0} D_u^{-j-1} C_u^* z_{t+j}$ (forward)
4. $s_0 = P_{ks}^{-1}(k_0 - P_{ku} u_0)$, then $s_{t+1} = D_s s_t + C_s^* z_t$ (backward)
5. $x_t = Px_t^*$. (map back)

- If $z_t = \bar{z}$ eventually, then $x_t \rightarrow \bar{x}$: the steady state is a **saddle**.
- The set of (k_0, y_0) with $y_0 = M_{uy}^{-1}(u_0 - M_{uk} k_0)$ is an n_k -dimensional subspace of $\mathbb{R}^n$: the **stable manifold** (stable arm, saddle path). The economy *jumps* onto it at $t = 0$ and rides it to the steady state.

[Miao Corollary 1.4.1, eq. (1.34)]

What the BK conditions mean

- **Counting.** Each unstable eigenvalue delivers one forward-looking restriction (a row of u_t); each jump variable needs exactly one restriction to pin down its starting value.
 - $n_u = n_y$: one restriction per free variable — **just right**.
 - $n_u > n_y$: too many restrictions — **no** stable solution.
 - $n_u < n_y$: too few — **infinitely many** stable solutions (indeterminacy).
- **Rank.** The n_u restrictions must involve the y 's independently (M_{uy} invertible); otherwise some jump variable is left free even when the count is right.
- **Hyperbolicity.** A unit root gives neither convergence nor explosion: the solution need not be stable.

Theorem (BK violated)

$n_u > n_y$: no stable solution. $n_u < n_y$: infinitely many. (Proof: $u_0 = M_{uk}k_0 + M_{uy}y_0$: n_u equations, n_y unknowns.)

[Miao Theorem 1.4.3]

Sanity check: the scalar cases are special cases

Equation	n_k	n_y	n_u	BK verdict
$x_{t+1} = bx_t + cz_t$, x_0 given, $ b < 1$	1	0	0	unique (backward)
$x_{t+1} = bx_t + cz_t$, x_0 free, $ b > 1$	0	1	1	unique (forward)
$x_{t+1} = bx_t + cz_t$, x_0 free, $ b < 1$	0	1	0	indeterminate
$x_{t+1} = bx_t + cz_t$, x_0 given, $ b > 1$	1	0	1	no stable solution
$x_{t+2} = ax_{t+1} + bx_t + cz_t$, saddle	1	1	1	unique (saddle path)

Everything from §1.1 and §1.3 is the $n \leq 2$ corner of one theorem.

Generalization 1: W not diagonalizable $\Rightarrow$ Jordan form

Diagonalizability can fail when eigenvalues repeat. Every square matrix still has a **Jordan decomposition** $W = PJP^{-1}$ with

$$J = \begin{bmatrix} J_1 & & \\ & \ddots & \\ & & J_l \end{bmatrix}, \quad J_i = \begin{bmatrix} \lambda_i & 1 & & \\ & \lambda_i & \ddots & \\ & & \ddots & 1 \\ & & & \lambda_i \end{bmatrix} \quad (m_i \times m_i).$$

- J is block-diagonal and upper triangular; when all blocks are 1×1 , $J = D$.
- The proof goes through unchanged: partition J into stable and unstable blocks, solve u forward, s backward. Powers J_i^t contain terms like $t\lambda_i^{t-1}$, which still vanish when $|\lambda_i| < 1$.
- This is how Miao and Blanchard–Kahn state the theorem. Numerically, Jordan forms are unstable to compute — software does not use them.

[Miao §1.4.1]

Generalization 2: A singular $\Rightarrow$ Klein and the QZ decomposition

A is singular whenever the model contains *intratemporal* equations (no $t+1$ variable). Example, Cagan: $R_t = p_{t+1} - p_t$, $m_t - p_t = -\alpha R_t$:

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} R_{t+1} \\ p_{t+1} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ \alpha & -1 \end{bmatrix} \begin{bmatrix} R_t \\ p_t \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} m_t.$$

We cannot form $A^{-1}B$. Klein (2000) instead uses the **generalized Schur (QZ) decomposition**:

$$QAZ = S, \quad QBZ = T, \quad Q, Z \text{ unitary, } S, T \text{ upper triangular,}$$

with **generalized eigenvalues** $\lambda_i = t_{ii}/s_{ii}$ ($s_{ii} = 0$ counts as ∞ : unstable).

- Stable pairs first; $x_t^* = Z^H x_t$. Then $Sx_{t+1}^* = Tx_t^* + QCz_t$ is block *triangular*: the unstable block is self-contained — forward; then the stable block — backward.
- BK conditions become $n_s = n_k$ and Z_{11} invertible. QZ always exists and is numerically robust: this is what **Dynare** uses.

[Miao §1.4.2, Theorems 1.4.4–1.4.5]

Generalization 3: Sims (gensys) — let the algebra find the jump variables

Sims (2002) writes the model as

$$\Gamma_0 y_t = \Gamma_1 y_{t-1} + \Psi \varepsilon_t + \Pi \eta_t,$$

where η_t are *endogenous* expectational errors ($\mathbb{E}_{t-1}\eta_t = 0$), e.g. $\eta_t = \pi_t - \mathbb{E}_{t-1}\pi_t$.

- No need to declare predetermined variables: every variable is dated when it is known, and the expectational errors carry the “jump”.
- QZ on (Γ_0, Γ_1) ; the unstable block must be zero, giving n_u linear restrictions on η_t .
- A **singular value decomposition** of the matrix multiplying η_t checks them: existence $\Leftrightarrow$ the restrictions can be met; uniqueness $\Leftrightarrow$ they pin down η_t completely. Otherwise: sunspots.

Same idea, three decompositions

BK: eigendecomposition of $A^{-1}B$. Klein: QZ of (A, B) . Sims: QZ of $(\Gamma_0, \Gamma_1) + \text{SVD}$.

In each case: **unstable directions forward, stable directions backward, count and rank.**

[Miao §2.3.3]

Phase diagrams: seeing the saddle

A graphical device for two-dimensional systems, linear or nonlinear. Take

$$x_{t+1} = \tilde{a}x_t + by_t - e, \quad y_{t+1} = cx_t - \tilde{d}y_t + f, \quad x_0 \text{ given},$$

and write it in *changes*, $\Delta x_t = x_{t+1} - x_t$, $\Delta y_t = y_{t+1} - y_t$:

$$\Delta x_t = ax_t + by_t - e, \quad \Delta y_t = cx_t - dy_t + f, \quad a = \tilde{a} - 1, \quad d = \tilde{d} + 1,$$

with $a, b, c, d, e, f > 0$.

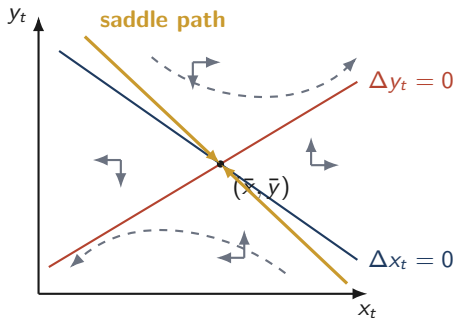
- **Isoclines:** the loci where one variable does not move.

$$\Delta x_t = 0 \iff y_t = -\frac{a}{b}x_t + \frac{e}{b} \quad (\text{downward sloping}), \quad \Delta y_t = 0 \iff y_t = \frac{c}{d}x_t + \frac{f}{d} \quad (\text{upward sloping})$$

- **Vector field:** $\Delta x_t > 0$ above the $\Delta x = 0$ line (x moves right); $\Delta y_t > 0$ below the $\Delta y = 0$ line (y moves up).
- Their intersection is the steady state.

[Miao §1.5]

Phase diagram



- North and south: arrows point toward the steady state in one direction, away in the other. Only paths starting on the **saddle path** converge.
- East and west: both arrows point away; every path diverges.
- x_0 given, y_0 free: the system **jumps** onto the saddle path above x_0 , then converges. The jump is the forward-looking solution.
- The saddle path is the stable manifold $y_0 = M_{uy}^{-1}(u_0 - M_{uk}k_0)$ in two dimensions.

End of the deterministic world.

Next: the same machinery when the future is uncertain.

Adding uncertainty

Fix a probability space and a **filtration** $\{\mathcal{F}_t\}_{t \geq 0}$: $\mathcal{F}_t$ is the information available at date t , and $\mathcal{F}_0 \subset \mathcal{F}_1 \subset \dots$ (we never forget).

- A **stochastic process** $\{X_t\}$ is a sequence of random variables. It is **adapted** if X_t is known at t (measurable w.r.t. $\mathcal{F}_t$). We assume all processes are adapted.
- $\mathbb{E}_t[\cdot] \equiv \mathbb{E}[\cdot \mid \mathcal{F}_t]$ is the conditional expectation given date- t information.

The exogenous shocks

$$z_{t+1} = \Phi z_t + \Sigma \varepsilon_{t+1}, \quad z_0 \text{ given}, \quad \mathbb{E}_t \varepsilon_{t+1} = 0, \quad \mathbb{E}_t \varepsilon_{t+1} \varepsilon'_{t+1} = I.$$

Φ is $n_z \times n_z$, Σ is $n_z \times n_\varepsilon$. Almost any linear shock process (AR(p), ARMA, VAR) can be stacked into this form.

[Miao §2.1, eq. (2.1)–(2.2)]

Covariance stationarity

Definition

$\{X_t\}$ is **covariance stationary** if it has constant mean $m = \mathbb{E}X_t$, finite second moments, and a covariance $\mathbb{E}[(X_t - m)(X_s - m)']$ that depends only on $|t - s|$.

For $z_{t+1} = \Phi z_t + \Sigma \varepsilon_{t+1}$ with all eigenvalues of Φ inside the unit circle:

- Mean: $\mu_{t+1} = \Phi \mu_t \rightarrow 0$. Start at $\mu_0 = 0$ and it stays there.
- Variance solves the discrete **Lyapunov equation**
 $C_z(0) \equiv \mathbb{E}z_t z_t' = \Phi C_z(0) \Phi' + \Sigma \Sigma'$.
- Autocovariances: $C_z(j) \equiv \mathbb{E}z_{t+j} z_t' = \Phi^j C_z(0)$.
- **Impulse response**: $z_{t+1} = \sum_{j \geq 0} \Phi^j \Sigma \varepsilon_{t+1-j}$, so $\Phi^j \Sigma$ is the effect of a shock today on z j periods later.

Stable $\Phi \Rightarrow \{z_t\}$ is a stable process: exactly the standing assumption we need. [Mish Def. 2.1.1]

Lag operators and expectations

We now need the lag operator to act on *expected* variables. Convention:

$$L^j X_t = X_{t-j}, \quad L^j \mathbb{E}_t X_{t+i} = \mathbb{E}_t X_{t+i-j}, \quad L^{-j} X_t = \mathbb{E}_t X_{t+j} \quad (i, j \geq 0).$$

- L shifts the *date of the variable*, not the *date of the information set*.
- Consistency: $L^j \mathbb{E}_t X_{t+j} = \mathbb{E}_t X_t = X_t$, which is why the lead operator must produce $\mathbb{E}_t X_{t+j}$ rather than X_{t+j} .
- Law of iterated expectations, $\mathbb{E}_t \mathbb{E}_{t+1} X = \mathbb{E}_t X$, is what makes repeated forward substitution work: $\mathbb{E}_t [\mathbb{E}_{t+1} X_{t+2}] = \mathbb{E}_t X_{t+2}$.

Rational expectations

Agents' forecasts are the model's conditional expectations $\mathbb{E}_t$. In the deterministic case this was perfect foresight; now it means *no systematic forecast errors*.

[Miao §2.2.1]

Scalar RE model: forward solution

$$\mathbb{E}_t x_{t+1} = b x_t + c z_t, \quad |b| > 1, \quad \{z_t\} \text{ bounded.}$$

In operator form $(L^{-1} - b) x_t = c z_t$, and expanding forward exactly as in §1.2,

$$x_t = \frac{c z_t}{L^{-1} - b} = -c \sum_{j=0}^{\infty} b^{-j-1} L^{-j} z_t = -c \sum_{j=0}^{\infty} b^{-j-1} \mathbb{E}_t z_{t+j}. \quad (\text{fundamental solution})$$

- Same formula as before with z_{t+j} replaced by $\mathbb{E}_t z_{t+j}$: a first glimpse of **certainty equivalence**.
- General solution: $x_t = x_t^* + B_t$ with $\mathbb{E}_t B_{t+1} = b B_t$ (a bubble).
- The stochastic **transversality condition** $\lim_{T \rightarrow \infty} b^{-T} \mathbb{E}_t x_{t+T} = 0$ rules bubbles out.

[Miao eq. (2.3)–(2.5)]

Scalar RE model: second order

$$\mathbb{E}_t x_{t+1} = a x_t + b x_{t-1} + c z_t, \quad x_0 \text{ given.}$$

Operator form: $(L^{-1} - a - bL) x_t = c z_t$. With characteristic roots $\lambda_1 + \lambda_2 = a$, $\lambda_1 \lambda_2 = -b$ and $|\lambda_1| < 1 < |\lambda_2|$:

$$(L^{-1} - \lambda_1)(L^{-1} - \lambda_2) L x_t = c z_t \quad \Rightarrow$$

$$x_t = \lambda_1 x_{t-1} - \frac{c}{\lambda_2} \sum_{j=0}^{\infty} \lambda_2^{-j} \mathbb{E}_t z_{t+j}$$

- Identical to the deterministic saddle-path solution of §1.3 with $z_{t+j} \rightarrow \mathbb{E}_t z_{t+j}$.
- Backward part (stable root λ_1) + forward part (unstable root λ_2).

[Miao eq. (2.6)–(2.9)]

The method of undetermined coefficients

Same equation, $\mathbb{E}_t x_{t+1} = ax_t + bx_{t-1} + cz_t$, now with $z_t = \rho z_{t-1} + \varepsilon_t$, $|\rho| < 1$.

1. **Guess** a linear solution in the state variables: $x_t = Gx_{t-1} + Hz_t$.
2. **Substitute.** Then $\mathbb{E}_t x_{t+1} = Gx_t + H\mathbb{E}_t z_{t+1} = G(Gx_{t-1} + Hz_t) + H\rho z_t$, so

$$G^2 x_{t-1} + (GH + \rho H) z_t = a(Gx_{t-1} + Hz_t) + bx_{t-1} + cz_t.$$

3. **Match coefficients** on x_{t-1} and on z_t :

$$G^2 = aG + b, \quad GH + \rho H = aH + c \Rightarrow H = \frac{c}{a + \rho - G}.$$

- The G equation is the characteristic equation: two roots. For a stable solution pick $|G| < 1$, i.e. $G = \lambda_1$. Then H follows.
- This is Uhlig's method in miniature; in matrix form the G equation becomes a *matrix quadratic*, and picking the stable root is the hard part.

[Miao §2.2.2]

Multivariate linear RE models

$$A \mathbb{E}_t x_{t+1} = B x_t + C z_t,$$

$x_t \in \mathbb{R}^n$, A, B are $n \times n$, C is $n \times n_z$, $\{z_t\}$ exogenous.

- **Stability** of a stochastic process: $\{X_t\}$ is stable if $\|X_t\|_{\max} < M$ for all t , where now $\|x\|_{\max} = \max_j \mathbb{E} |x_j|$. We assume $\{z_t\}$ is stable and look for a stable $\{x_t\}$.
- **Predetermined** (Blanchard–Kahn): $\{X_t\}$ is predetermined if X_0 is given and X_{t+1} is known at t : $\mathbb{E}_t X_{t+1} = X_{t+1}$.
(Klein generalizes this: $X_{t+1} - \mathbb{E}_t X_{t+1}$ may be an exogenous, mean-zero surprise.)
- Order $x_t = [k'_t, y'_t]'$ as before: n_k predetermined, n_y non-predetermined.

Two issues, three methods

(a) A may be singular; (b) some variables are predetermined, others are not. BK (needs A invertible), Klein (QZ), and Sims (QZ + SVD) handle these differently — as in ch. 1.

[Miao §2.3, Def. 2.3.1]

Blanchard–Kahn, stochastic version

With A invertible: $\mathbb{E}_t x_{t+1} = Wx_t + A^{-1}Cz_t$, $W = PDP^{-1}$, $x_t^* = P^{-1}x_t$, so

$$\mathbb{E}_t x_{t+1}^* = D x_t^* + C^* z_t.$$

- **Step 1** (forward): $u_t = - \sum_{j=0}^{\infty} D_u^{-j-1} C_u^* \mathbb{E}_t z_{t+j}$.
- **Step 2**: $y_t = M_{uy}^{-1}(u_t - M_{uk} k_t)$ — unchanged.
- **Step 3** (backward): k_t predetermined means $\mathbb{E}_t k_{t+1} = k_{t+1}$; the stable block iterates as before.

Certainty equivalence (BK conditions carry over verbatim)

The stochastic solution is the deterministic solution with every future z_{t+j} replaced by $\mathbb{E}_t z_{t+j}$. Only the *conditional mean* of the shocks matters; their variance does not affect decision rules. (A property of linear models; no precautionary behaviour, no risk premia.)

[Miao §2.3.1, eq. (2.12)]

What the solution looks like

If $z_{t+1} = \Phi z_t + \Sigma \varepsilon_{t+1}$ with Φ stable, then $\mathbb{E}_t z_{t+j} = \Phi^j z_t$, so the forward sum collapses:

$$u_t = - \sum_{j=0}^{\infty} D_u^{-j-1} C_u^* \Phi^j z_t \equiv N z_t \quad (N \text{ solves a Sylvester equation: } D_u N - N \Phi = -C_u^*).$$

Substituting through steps 2–3, the whole solution is a pair of *linear policy functions*:

$$y_t = A_{yk} k_t + A_{yz} z_t, \quad k_{t+1} = A_{kk} k_t + A_{kz} z_t$$

- Jump variables are a function of the current *state* (k_t, z_t) ; states evolve with a stable A_{kk} .
- This is exactly what Dynare prints as the “policy and transition functions”.
Everything we do from Session 5 onward produces objects of this form.

[Miao §2.3.2, solution form]

Example: an interest-rate rule in a New Keynesian model

A log-linearized three-equation NK model:

$$\pi_t = \kappa x_t + \beta \mathbb{E}_t \pi_{t+1}$$

Phillips curve (AS)

$$x_t = \mathbb{E}_t x_{t+1} - \frac{1}{\gamma} (i_t - \mathbb{E}_t \pi_{t+1}) + \varepsilon_t^x$$

IS curve (AD)

$$i_t = \rho_r i_{t-1} + \varepsilon_t$$

policy rule, $|\rho_r| < 1$

π_t inflation, x_t output gap, i_t nominal rate; $\kappa, \beta, \gamma > 0$; $\varepsilon_t, \varepsilon_t^x$ IID.

- The central bank sets the rate *exogenously*: it smooths past rates and does not react to inflation or output.
- Question: does this economy have a unique stable equilibrium?
- **Classify.** i_{t-1} is predetermined ($n_k = 1$). x_t and π_t are jump variables ($n_y = 2$). So BK needs **exactly two** unstable eigenvalues.

[Miao Example 2.3.1, eq. (2.13)–(2.15)]

Step 0: write it as $A\mathbb{E}_t x_{t+1} = Bx_t + Cz_t$

Put every $t+1$ (expected) variable on the left, everything dated t or earlier on the right.
State vector $x_t = [i_{t-1}, x_t, \pi_t]'$:

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{\gamma} & 1 & \frac{1}{\gamma} \\ 0 & 0 & \beta \end{bmatrix}}_A \begin{bmatrix} i_t \\ \mathbb{E}_t x_{t+1} \\ \mathbb{E}_t \pi_{t+1} \end{bmatrix} = \underbrace{\begin{bmatrix} \rho_r & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -\kappa & 1 \end{bmatrix}}_B \begin{bmatrix} i_{t-1} \\ x_t \\ \pi_t \end{bmatrix} + \begin{bmatrix} \varepsilon_t \\ -\varepsilon_t^x \\ 0 \end{bmatrix}.$$

- Row 2 is the IS curve, row 3 the Phillips curve.
- i_t sits on the left as “ $\mathbb{E}_t i_t$ ”: known at t , so predetermined for $t+1$, as BK requires.
- $\det A = \beta \neq 0$: A is invertible, so BK applies directly.

[Miao eq. (2.16)]

Step 1: $W = A^{-1}B$ and its eigenvalues

$$W = A^{-1}B = \begin{bmatrix} \rho_r & 0 & 0 \\ \frac{\rho_r}{\gamma} & 1 + \frac{\kappa}{\beta\gamma} & -\frac{1}{\beta\gamma} \\ 0 & -\frac{\kappa}{\beta} & \frac{1}{\beta} \end{bmatrix}$$

- Block lower-triangular: one eigenvalue is $\lambda_1 = \rho_r$, with $|\rho_r| < 1$: **stable**.
- The other two are the eigenvalues of the lower-right 2×2 block W_{22} :

$$\text{tr } W_{22} = 1 + \frac{1}{\beta} + \frac{\kappa}{\beta\gamma}, \quad \det W_{22} = \frac{1}{\beta}.$$

- Characteristic polynomial $p(\lambda) = \lambda^2 - (\text{tr } W_{22})\lambda + \det W_{22}$. Evaluate at $\lambda = 1$:

$$p(1) = 1 - \text{tr } W_{22} + \det W_{22} = -\frac{\kappa}{\beta\gamma} < 0.$$

Step 2: count unstable eigenvalues

- $p(1) < 0$ with $p(\lambda) \rightarrow +\infty$ as $\lambda \rightarrow \pm\infty$: the parabola crosses zero once below 1 and once above 1. Both roots real; $\lambda_2 < 1 < \lambda_3$.
- Both positive since their product is $1/\beta > 0$ and their sum is positive. So $0 < \lambda_2 < 1$: stable.
- Numerically, with $\beta = 0.99$, $\kappa = 0.1$, $\gamma = 1$, $\rho_r = 0.5$: $\lambda \in \{0.50, 0.73, 1.38\}$.

	required	actual
unstable eigenvalues n_u	$= n_y = 2$	1
stable eigenvalues n_s	$= n_k = 1$	2

Verdict: $n_u < n_y \Rightarrow$ indeterminacy

Infinitely many stable equilibria. Any (x_0, π_0) on a one-dimensional family of starting points is consistent with the model; sunspots can drive inflation.

What the example teaches

- An interest-rate rule that does *not respond to endogenous variables* leaves inflation expectations unanchored. The math says so before any economics is invoked: one unstable root is not enough to tie down two jump variables.
- Exercise (Miao ch. 2, Ex. 2): let $i_t = \rho_r i_{t-1} + \phi_\pi \pi_t + \phi_x x_t + \varepsilon_t$. Find the condition on ϕ_π (and ϕ_x) that delivers exactly two unstable roots. You will rediscover the **Taylor principle**.
- Notice what we did *not* need: the variances of the shocks, or the distribution of ε . Certainty equivalence at work.
- And what we did need: (1) write the model in $A\mathbb{E}_t x_{t+1} = Bx_t + Cz_t$ form; (2) classify variables; (3) count eigenvalues; (4) check rank. **This is the routine.**

Summary

Ideas

- Two questions: is the initial value given? is the root inside the unit circle?
- Stable roots: backward from initial conditions. Unstable roots: forward from transversality conditions.
- Lag operators make this mechanical: expand backward when $|\lambda| < 1$, forward when $|\lambda| > 1$.
- Diagonalize to decouple a system into scalar problems; count and rank for existence and uniqueness.

Results

- **Blanchard–Kahn:** $n_u = n_y$, M_{uy} invertible, no unit roots $\Rightarrow$ unique stable solution.
- $n_u > n_y$: none. $n_u < n_y$: infinitely many.
- Jordan, Klein (QZ), Sims (QZ + SVD) extend the same logic.
- Stochastic case: same solution with $\mathbb{E}_t$; certainty equivalence.
- Output: linear policy functions
$$y_t = A_{yk}k_t + A_{yz}z_t,$$
$$k_{t+1} = A_{kk}k_t + A_{kz}z_t.$$

Reading and what comes next

- **Reading:** Miao (2020), *Economic Dynamics in Discrete Time*, §1.1–1.5 and §2.1–2.3. Original sources: Blanchard and Kahn (1980, *Econometrica*); Klein (2000, *JEDC*); Sims (2002, *Computational Economics*); Uhlig (1999).
- **Problem set 1** (analytical) will ask you to: solve scalar equations forward and backward; apply BK by hand to a 2×2 and a 3×3 system; and derive the Taylor-principle condition in the NK example.
- **Next sessions:** the Klein and Sims methods in more detail and the QZ decomposition; then where the linear system comes from — steady states and linearization of a nonlinear model.

Unstable forward, stable backward.

Count them, check the rank, and you have solved the model.